

Tutorial 1

$$1.4) I = \frac{Q}{t} = \frac{300}{30} = \underline{10A}$$

$$1.6) I = \frac{dq}{dt}$$

$$a.) t = 1ms \quad I = \frac{30-0}{2-0} = \underline{15A}$$

$$b.) t = 6ms \quad I = \frac{30-30}{6-2} = \underline{0A} \quad (\text{no change in charge})$$

$$c.) t = 10ms \quad I = \frac{0-30}{12-8} = \underline{-7.5A} \quad (\text{negative sign shows change of current direction})$$

$$1.8) q = \int i dt = \frac{10 \times 1}{2} + 10 \times 1 = 15 \mu A$$

$$1.11) I = 90mA$$

$$t = 12h = 12 \times 60 \times 60 = 43200s$$

$$Q = I \cdot t = (90 \times 10^{-3}) \cdot (43200) \\ = 3888 C$$

$$E = P \cdot t = I V t = Q \cdot V$$

$$= 3.888 k \times 1.5 = 5.832 kJ$$

1.12) ① For $0 < t < 6s$ assuming $q(0) = 0$

$$q(t) = \int_0^t i dt + q(0) = \int_0^t 3t dt = 1.5 t^2$$

$$\therefore \text{at } t=6 \quad q(6) = 1.5 (6)^2 = 54$$

② For $6s < t < 10s$

$$\begin{aligned} q(t) &= \int_6^t i dt + q(6) = \int_6^t 18 dt + 54 \\ &= 18t - 54 \end{aligned}$$

$$\therefore \text{at } t=10 \quad q(10) = 180 - 54 = 126$$

③ For $10s < t < 15s$

$$\begin{aligned} q(t) &= \int_{10}^t i dt + q(10) = \int_{10}^t (-12) dt + 126 \\ &= -12t + 246 \end{aligned}$$

$$\therefore \text{at } t=15 \quad q(15) = 66$$

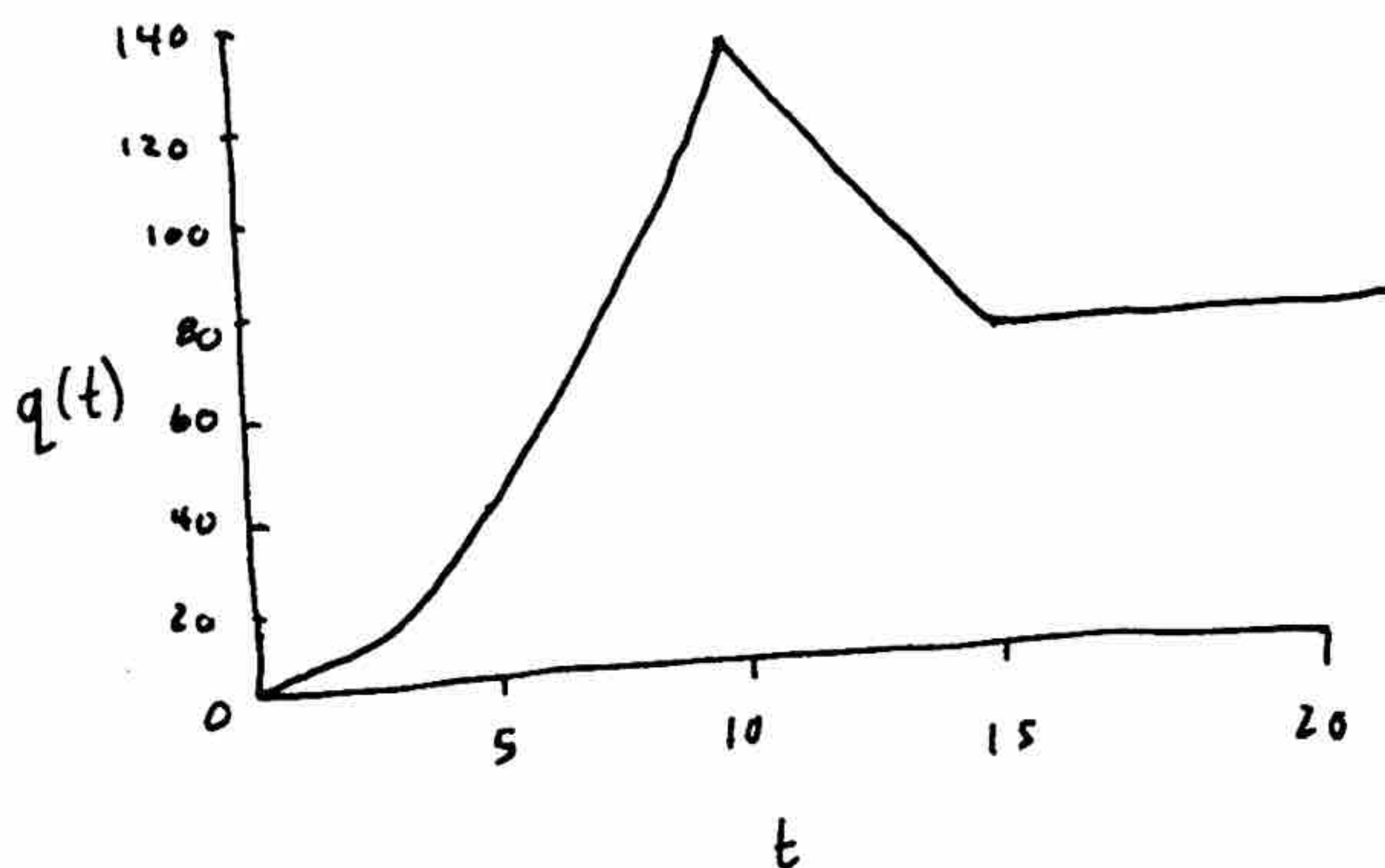
④ For $15s < t < 20s$

$$q(t) = \int_{15}^t 0 dt + q(15) = 66$$

∴

$$q(t) = \begin{cases} 1.5t^2 \text{ C}, & 0 < t < 6s \\ 18t - 54 \text{ C}, & 6s < t < 10s \\ -12t + 246 \text{ C}, & 10s < t < 15s \\ 66 \text{ C}, & 15s < t < 20s \end{cases}$$

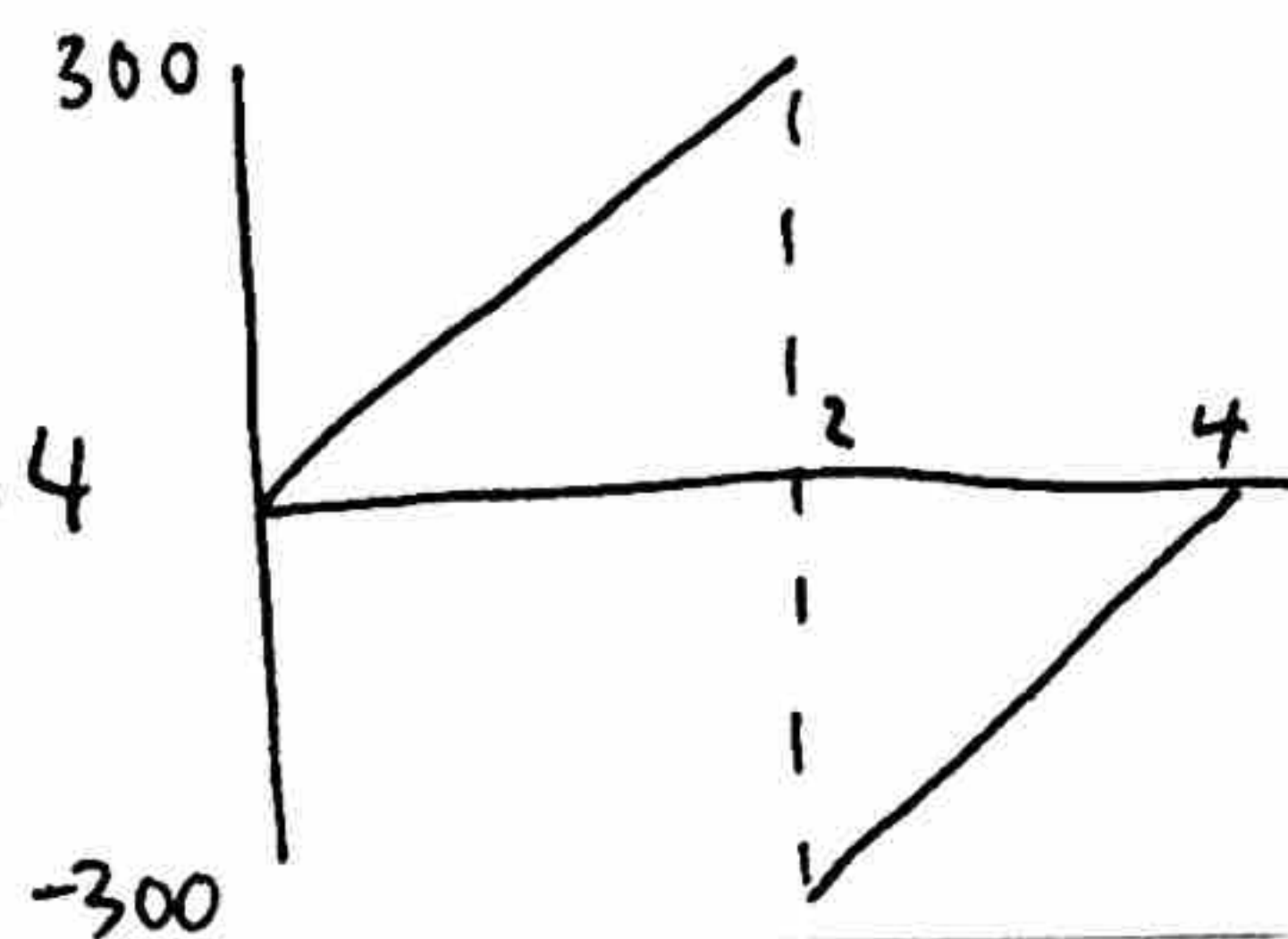
∴ plot $q(t)$



1.16) a) $i(t) = \begin{cases} 30t \text{ mA}, & 0 < t < 2 \\ 120 - 30t \text{ mA}, & 2 < t < 4 \end{cases}$

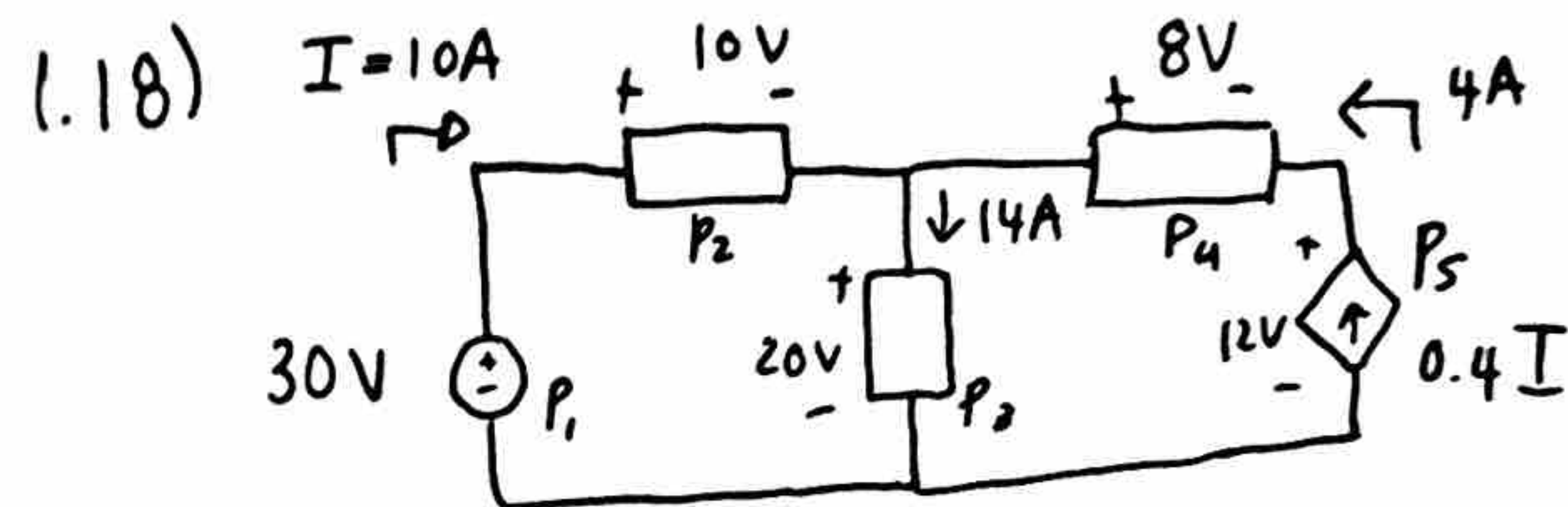
$$v(t) = \begin{cases} 5V, & 0 < t < 2 \\ -5V, & 2 < t < 4 \end{cases}$$

$$\therefore p(t) = i \cdot v = \begin{cases} 150t \text{ mW}, & 0 < t < 2 \\ -600 + 150t \text{ mW}, & 2 < t < 4 \end{cases}$$



b) From the graph p

$$W = \int_0^4 p \, dt = 0 \text{ J}$$



$$P = VI$$

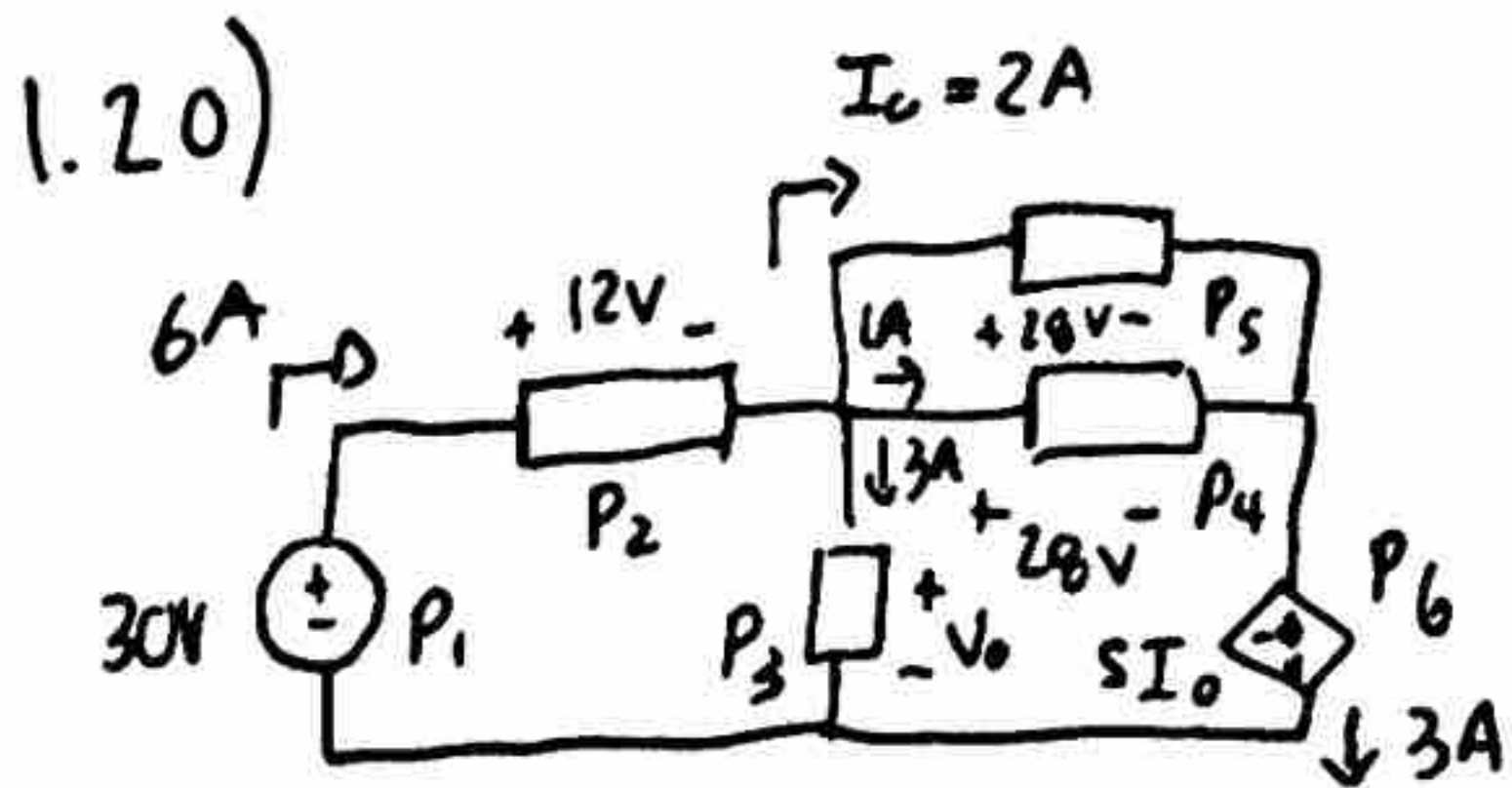
$$\therefore P_1 = 30(-10) = -300W$$

$$P_2 = 10(10) = 100 \text{ W}$$

$$P_3 = 20(14) = 280 \text{ W}$$

$$P_4 = 8(-4) = -32 \text{ W}$$

$$P_s = 12(-0.4 \times 10) = -48 \text{ W}$$



$$P = VI$$

$$\therefore V_0 = \frac{P_3}{3} = \frac{54}{3} = 18 \text{ V}$$

$$\therefore P_1 = 30(-6) = -180 \text{ W}$$

$$P_2 = 12(6) = 72 \text{ W}$$

$$P_3 = V_0(3) = ?$$

$$P_4 = 28(1) = 28W$$

$$P_s = 28(2) = 56 \text{ W}$$

$$P_1 = \epsilon(\gamma, \gamma) = -2n|a|$$

$$\sum p = 0$$

$$\therefore -180 + 72 + p3 + 28 + 56 - 30 = 0$$

$$\therefore P_3 = 54 \text{ W}$$

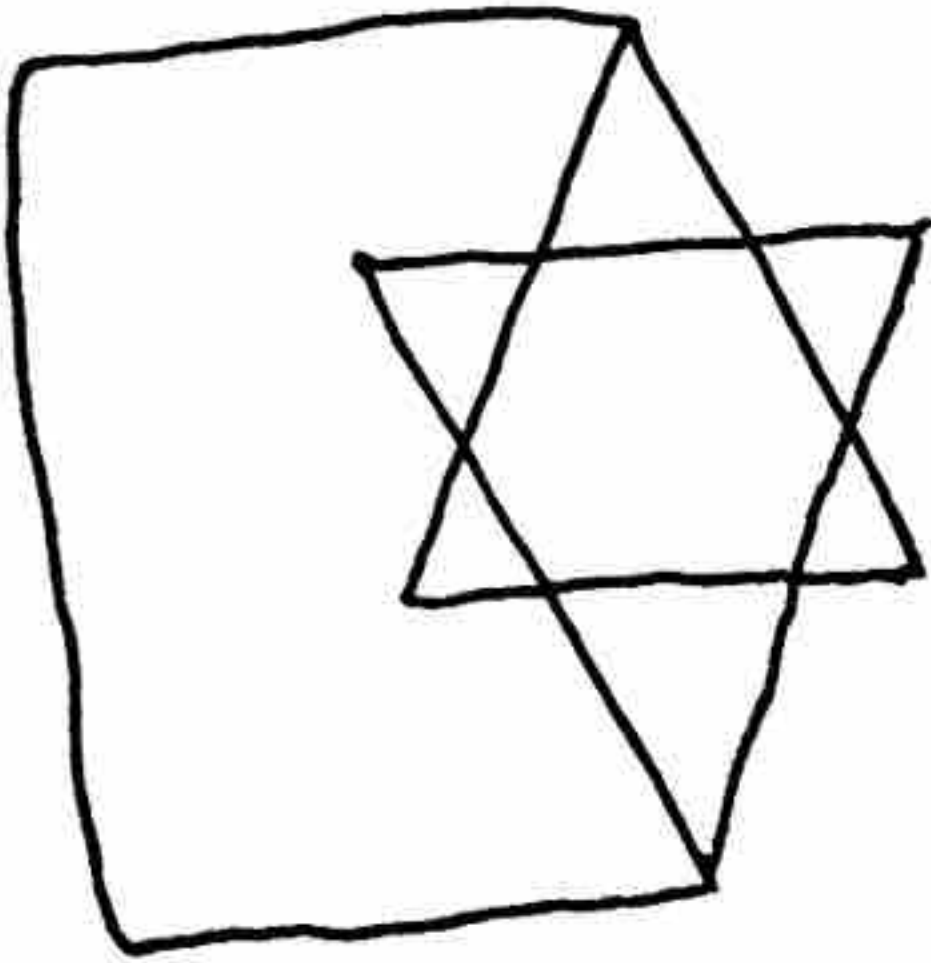
$$1.24) W = pt = 60 \times 24 \text{ wh} = 1,44 \text{ Kwh}$$

$$C = 8.2 \text{ cents} \times 1,44 = 11,808 \text{ cents}$$

$$1.31) [(230 + 75) \times 6 \times 30] \div 1000 = 54,9 \text{ Kwh}$$

$$\therefore 11,75 \times 54,9 = 645,08 \text{ cents}$$

2.6)



$$n = 8$$

$$e = 8$$

$$b = n + e - 1 \\ = 15$$